

Explaintool HHGR Evaluation Report

dataset: contract_act_gold

document_id: 0940d367554383c5

top_k: 5

items_evaluated: 10

seed: 42

elapsed_seconds: 0.244

generated_at: 2026-08-05T17:10:44Z

Retrieval Accuracy & Latency by System

System	items	Recall@K	Precision@K	Hit Rate@K	MRR	MAP	NDCG@K	Mean Latency (ms)	p50 Latency (ms)	p95 Latency (ms)	Throughput (ops/sec)
Baseline	10	0.8000	0.1800	0.8000	0.7500	0.7500	0.7631	2.7510	2.5950	4.3880	363.4600
Baseline	10	0.8000	0.1800	0.8000	0.7000	0.6750	0.7139	0.4610	0.4440	0.5950	2168.2600
Baseline	10	0.9000	0.2000	0.9000	0.7583	0.7283	0.7781	0.0280	0.0250	0.0560	36363.6400
Baseline	10	0.8000	0.1800	0.8000	0.6500	0.6200	0.6743	0.2300	0.2210	0.4500	4357.3000
Baseline	10	0.8000	0.1800	0.8000	0.7000	0.6750	0.7139	0.5180	0.4800	0.7760	1929.0100

HHGR Explainability Metrics (per query)

item_id	Citation Accuracy	Hierarchy Correctness	Graph Path Accuracy	Provenance Completeness	Evidence Coverage	Counter-Authority F1	confidence
ica-001	1.0000	1.0000	1.0000	0.9500	0.9444	1.0000	0.8301
ica-002	1.0000	1.0000	1.0000	0.9500	1.0000	1.0000	0.7094
ica-003	1.0000	1.0000	1.0000	0.9643	0.6667	1.0000	0.7794
ica-004	1.0000	1.0000	1.0000	0.9643	0.5625	1.0000	0.7183
ica-005	1.0000	1.0000	1.0000	0.9643	0.7143	1.0000	0.7217
ica-006	1.0000	1.0000	1.0000	0.4214	0.0000	1.0000	0.4281
ica-007	0.5000	1.0000	1.0000	0.9643	0.8000	1.0000	0.7285
ica-008	1.0000	1.0000	1.0000	0.4500	0.5000	1.0000	0.6331
ica-009	1.0000	1.0000	1.0000	0.9500	1.0000	1.0000	0.7457
ica-010	1.0000	1.0000	1.0000	0.9500	0.6364	1.0000	0.4997

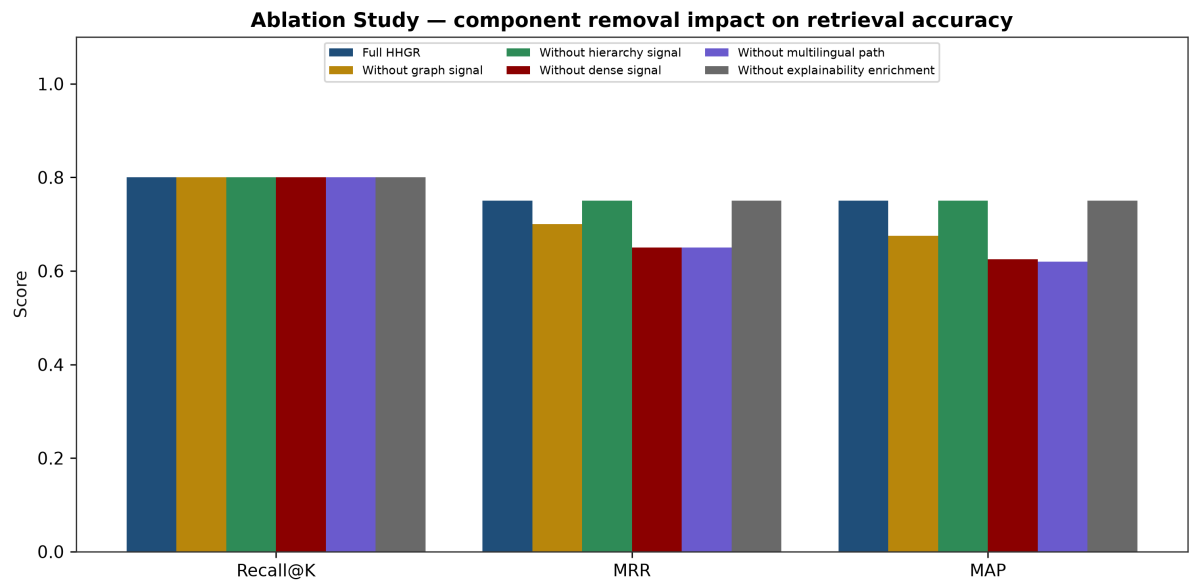
RAGAS-style Generation Metrics (offline surrogate)

system	item_id	Faithfulness	Answer Relevancy	Context Recall	Context Precision	Answer Correctness
hhgr	ica-001	1.0000	0.8081	1.0000	0.5731	0.5520
hhgr	ica-002	1.0000	0.7542	1.0000	0.4711	0.2549
hhgr	ica-003	1.0000	0.7864	1.0000	0.4804	0.4495
hhgr	ica-004	1.0000	0.7842	1.0000	0.4517	0.5843
hhgr	ica-005	1.0000	0.8563	1.0000	0.5079	0.3928

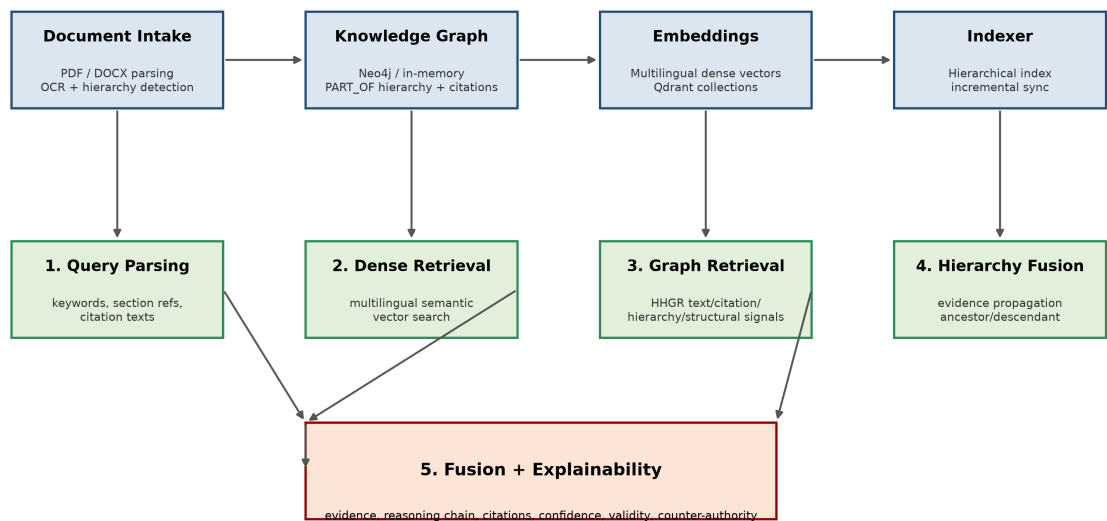
system	item_id	Faithfulness	Answer Relevancy	Context Recall	Context Precision	Answer Correctness
hhgr	ica-006	1.0000	0.6211	0.0000	0.0510	0.2331
hhgr	ica-007	1.0000	0.7548	0.5000	0.5376	0.4694
hhgr	ica-008	1.0000	0.8524	0.0000	0.4771	0.6262
hhgr	ica-009	1.0000	0.7178	1.0000	0.2542	0.4984
hhgr	ica-010	1.0000	0.8926	1.0000	0.3881	0.3366
naive_rag	ica-001	1.0000	0.8484	1.0000	0.5731	0.5714
naive_rag	ica-002	1.0000	0.8000	1.0000	0.4748	0.2748
naive_rag	ica-003	1.0000	0.8329	1.0000	0.4663	0.4634
naive_rag	ica-004	1.0000	0.8000	1.0000	0.4359	0.6186
naive_rag	ica-005	1.0000	0.8981	1.0000	0.5037	0.3976
naive_rag	ica-006	1.0000	0.6708	1.0000	0.0510	0.2518
naive_rag	ica-007	1.0000	0.8030	1.0000	0.5329	0.4534
naive_rag	ica-008	1.0000	0.8932	1.0000	0.4746	0.6672
naive_rag	ica-009	1.0000	0.7276	1.0000	0.2562	0.5192
naive_rag	ica-010	1.0000	0.9125	1.0000	0.3881	0.3455

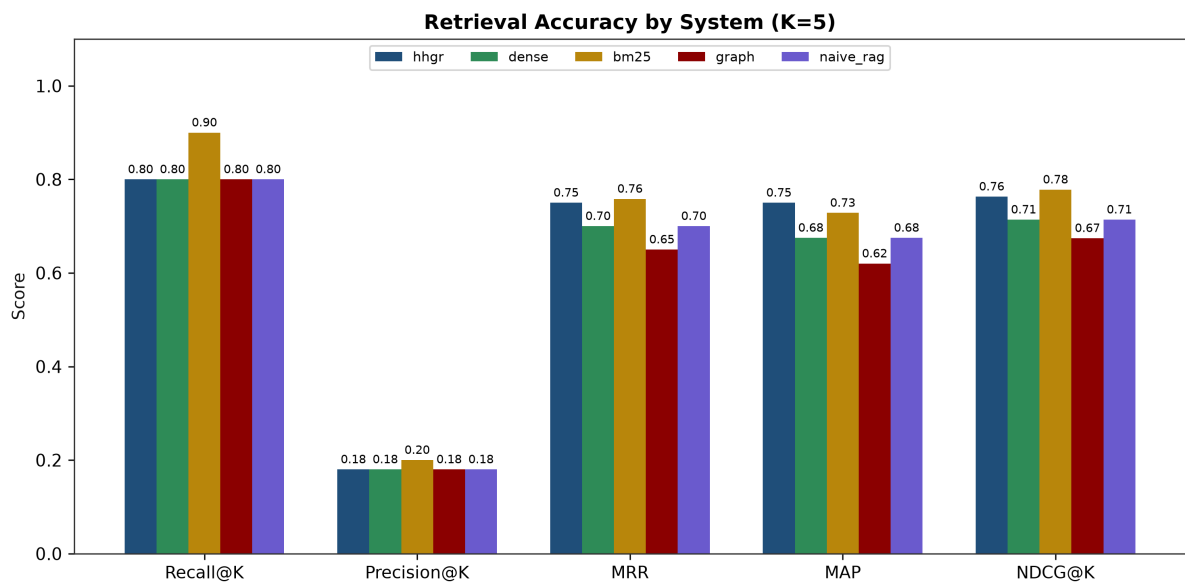
Ablation Study

ablation	label	Recall@K	MRR	MAP	NDCG@K	explain_citation_accuracy
full	Full HHGR	0.8000	0.7500	0.7500	0.7631	0.9500
no_graph	Without graph signal	0.8000	0.7000	0.6750	0.7139	0.9500
no_hierarchy	Without hierarchy signal	0.8000	0.7500	0.7500	0.7631	0.9500
no_dense	Without dense signal	0.8000	0.6500	0.6250	0.6770	0.9500
no_multilingual	Without multilingual path	0.8000	0.6500	0.6200	0.6743	0.7500
no_explainability	Without explainability enrichment	0.8000	0.7500	0.7500	0.7631	None

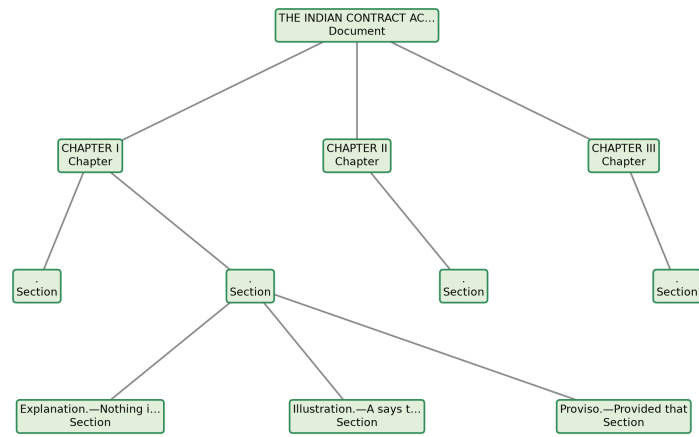


Explaintool: Explainable Multilingual Hierarchical Graph-RAG (HHGR) Architecture

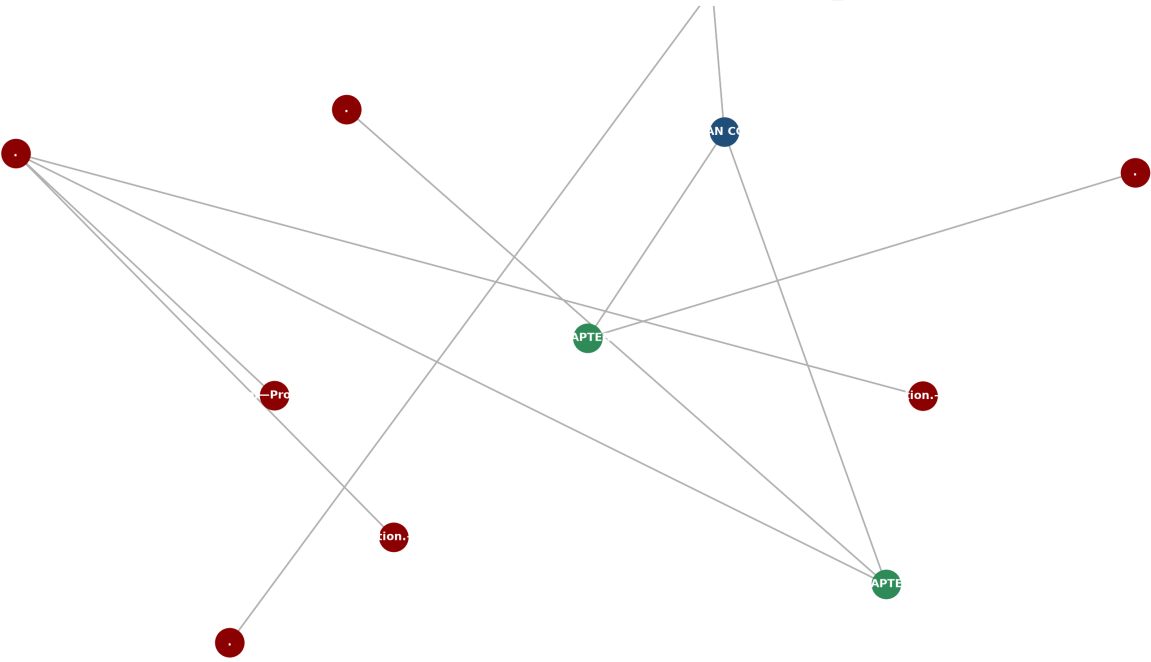


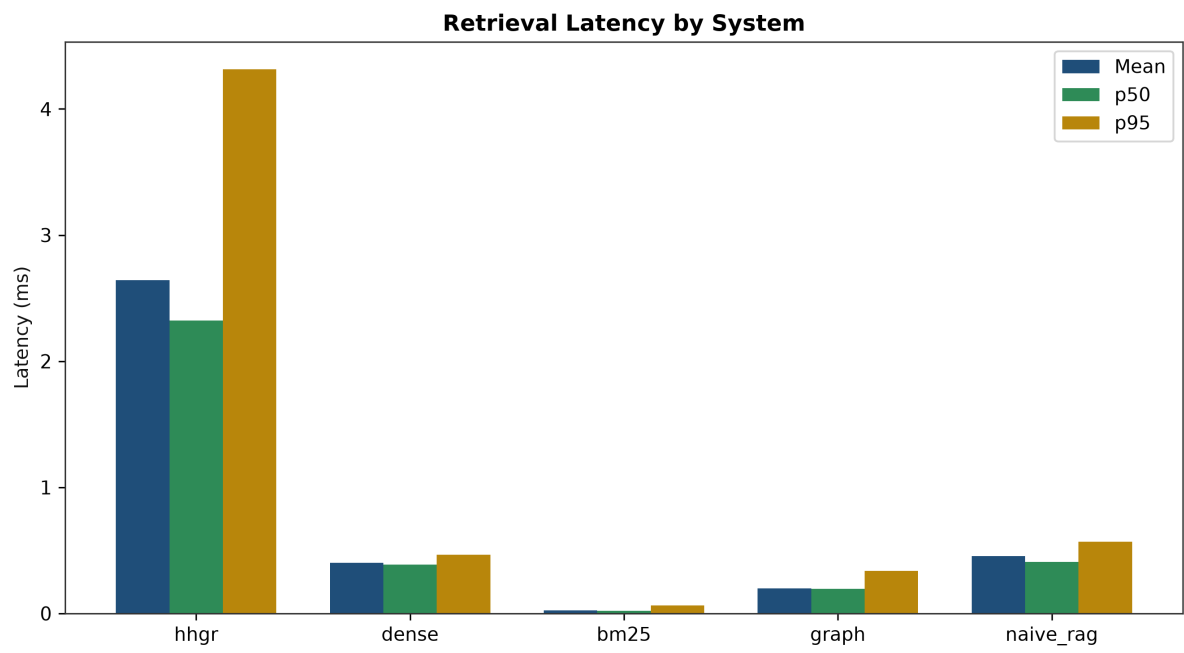


Hierarchy Tree — 0940d367554383c5.json



Knowledge Graph Example — Indian Contract Act, 1892 (PART_OF + citation edges)





RAGAS-style Generation Metrics (offline surrogate)

